Exact Enclosing-Area Transport and Sharp Penalty Budgets for the Spacetime Penrose Construction

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M518

October 9, 2026

Abstract

The source manuscript studies the spacetime Penrose inequality using the infimum over full enclosing cuts in the original metric. We give two local refinements of its comparison mechanism. A cutoff convex-gradient flow replaces the source's asymptotically lossless radial comparison by an exactly nondecreasing enclosing-area comparison whenever the stated core domination and fixed exterior metric lower bound hold. This result requires no minimizing cut, energy condition, or field equation. For the source's filled graph system, a one-variable convex minimization retains the physical penalty weights and gives a cubic-root formula for the worst adverse source contribution. We also derive the signed trace/divergence residual identity and its separate curvature and ADM budgets, with an explicit distinction between pointwise, integral, and weak residual control. All global constraint preparation, coupled elliptic solvability, detachment, and Riemannian Penrose inputs remain external to these deductions. Classical flow and convexity tools are credited; neither historical priority nor a new proof of the spacetime inequality is claimed. Contribution assessment: C3/5, exact full-cut transport and a source-weighted sharp penalty certificate are notable local refinements, conditional on the stated geometric and analytic inputs.

Contents

1 The geometric quantity and the source claim **2**

2 An exactly area-preserving end comparison **3**
2.1 A warped example and a necessary hypothesis 5

3 The actual filled-graph source and its measures **5**

4 A sharp pointwise penalty budget **6**

5 Trace, divergence and boundary residuals **8**

6 Finite-stage interpretation and usable area lower bounds **9**

7 Dependency map, comparison, and limitations **10**
7.1 Source architecture 10
7.2 What the Ram comparison contributes 11
7.3 Advantages, costs, and failure modes 11
7.4 Feasible extensions 12

8 Current literature and verification scope **12**

9 Conclusion **13**

1 The geometric quantity and the source claim

Throughout, $n \geq 3$, $k = n-1$, and $\omega_k = |\mathbb{S}^k|$. The smooth oriented exterior Ω has nonempty compact smooth boundary S, is connected and complete with its boundary included, and has one coordinate end with compact complement. The metric is g; K is a symmetric covariant two-tensor. We use the source's geometric constraint normalization

$$2\mu = R_g + (\operatorname{tr}_g K)^2 - |K|_g^2, \qquad J = \operatorname{div}_g\big(K - (\operatorname{tr}_g K)g\big). \tag{1.1}$$

Here J is a covector. The dominant energy condition (DEC) is $\mu \geq |J|_g$, with both densities integrable in dV_g. The boundary normal ν points into the exterior, toward the end:

$$\theta_+ = H_S + \operatorname{tr}_S K \leq 0, \qquad H_S = \operatorname{div}_S \nu.$$

There is no imposed sign on the other null expansion. The ADM conventions are

$$E = \frac{1}{2k\omega_k} \lim_{r\to\infty} \int_{S_r} (\partial_j g_{ij} - \partial_i g_{jj}) n_\delta^i \, \mathrm{d}A_\delta, \tag{1.2}$$

$$P_i = \frac{1}{k\omega_k} \lim_{r\to\infty} \int_{S_r} (K_{ij} - (\operatorname{tr}_g K) g_{ij}) n_\delta^j \, \mathrm{d}A_\delta. \tag{1.3}$$

When $E > |P|_\delta$, the invariant mass is $m = (E^2 - |P|_\delta^2)^{1/2}$. It is not the energy in an arbitrary asymptotic frame.

An admissible outer domain $D \subset \Omega$ is a connected smooth codimension-zero submanifold, closed in Ω, with interior in $\operatorname{int}\Omega$, containing the entire sufficiently distant end. Its entire intrinsic boundary $\Gamma = \partial D$ is compact, smooth, embedded and two-sided. Set

$$\mathsf{A}_g(S) = \inf_D \operatorname{Area}_g(\partial D). \tag{1.4}$$

All components and all boundary-coincident portions count; $D = \Omega$ is allowed. Neither attainment nor a trapped/minimal condition on the competitors is part of this definition. A finite list of candidate cuts gives upper bounds for this infimum, not lower bounds.

What is asserted upstream. The September 27 source [1] claims

$$m \geq \frac{1}{2}\left(\frac{\mathsf{A}_g(S)}{\omega_k}\right)^{(n-2)/(n-1)} \tag{1.5}$$

under its one-ended smooth data hypotheses, O_6/O_5 differentiated decay with $(n-2)/2 < q < n-2$, and positive enclosing area. It separately removes the initially assumed timelikeness. In dimensions

three and four it claims the O_2/O_1 regime $q > (n-2)/2$, including positive area and timelikeness as conclusions. These remain [S] claims here, not consequences of our finite checks. In the source's three-dimensional weak-decay section the symbols μ, J are physical densities: multiply both by 8π to recover (1.1). No such conversion is made in its four-dimensional section.

The all-dimensional numerical endpoint uses a companion theorem on possibly singular minimizing frontiers. This report neither audits that companion nor proves the source's global filled-system existence theorem. The distinction matters even though the scalar identity can be checked locally.

2 An exactly area-preserving end comparison

The source's strong end preparation, lines 573–627, and four-dimensional rest replacement, lines 747–790, compare all cuts using a radial compression with an $o(1)$ tensor loss. The following result removes that loss under the very metric bounds used there. The map is used to transport cuts, not to identify ADM energies in a rescaled coordinate frame.

Lemma 2.1 (A contracting end diffeomorphism)**.** *Suppose g is smooth and $g_{ij} - \delta_{ij} = O_1(r^{-q})$ on one coordinate end, for some $q > 0$. For every fixed $T > 0$ there is a smooth proper diffeomorphism $\Phi_T : \Omega \to \Omega$, isotopic to the identity and equal to it on a neighborhood of the compact core and S, such that*

$$\Phi_T^* g \le g, \qquad \Phi_T(x) = e^{-T}x + O_T(r^{1-q}), \qquad D\Phi_T = e^{-T}I + O_T(r^{-q}). \tag{2.1}$$

In particular $\Phi_T^* g = e^{-2T}\delta + O_T(r^{-q})$ in end coordinates.

Proof. On the end put $F = r^2/2$. Since $\Gamma(g) = O(r^{-1-q})$,

$$(\mathrm{Hess}_g F)_{ij} = \delta_{ij} - \Gamma^a_{ij} x_a = \delta_{ij} + O(r^{-q}).$$

Choose L so large that $\mathrm{Hess}_g F \ge \frac{1}{2}\delta$ on $r \ge L$. Choose a smooth nondecreasing $\chi : \mathbb{R} \to [0,1]$ that is zero for $F \le L^2/2$ and one for $F \ge 2L^2$. Define $X = \chi(F)\nabla_g F$ on the end and extend it by zero on the core. Flatness of the cutoff at its constant regions makes the extension smooth. The exact Lie derivative is

$$\mathcal{L}_X g = 2\chi(F)\,\mathrm{Hess}_g F + 2\chi'(F)\,\mathrm{d}F \otimes \mathrm{d}F \ge 0. \tag{2.2}$$

The field grows at most linearly in r, and is zero near S. The differential inequality $|\dot r| \le C(1+r)$ prevents finite-time escape; on compact sets ordinary smooth ODE continuation applies. Thus both time directions exist for every finite time. Let Φ_t be the flow of $-X$. Each Φ_t has inverse the time-t flow of X, hence is a diffeomorphism and proper. Differentiating the pullback gives

$$\frac{d}{dt}\Phi_t^* g = -\Phi_t^*(\mathcal{L}_X g) \le 0.$$

Integration proves the first assertion of (2.1).

For fixed T, sufficiently distant starting points remain on the region where $\chi = 1$ for $0 \le t \le T$. There

$$X(x) = x + O(r^{1-q}), \qquad DX(x) = I + O(r^{-q}).$$

The bound $|\dot r| \le Cr$ implies $r(\Phi_t x) \asymp_T r(x)$. Variation of constants in $\dot\Phi_t = -\Phi_t + O_T(r^{1-q})$ gives the position estimate. The variational equation $\dot J_t = -DX(\Phi_t x)J_t$, $J_0 = I$, and Gronwall's inequality give $J_t = e^{-t}I + O_T(r^{-q})$. Substitution in $J_T^{\mathsf{T}} g(\Phi_T x) J_T$ proves the pullback asymptotic. Only the original metric and a fixed T enter these constants. □

Theorem 2.2 (Lossless full-cut transfer)**.** *Assume the setting of Lemma 2.1. Suppose smooth positive metrics $\widehat{g}_R$, indexed by arbitrarily large R, satisfy*

$$\widehat{g}_R \geq g \quad \text{on the core and } r \leq R, \qquad \widehat{g}_R \geq c\delta \quad (r \geq R) \tag{2.3}$$

for a fixed $c > 0$. Then a single proper diffeomorphism Φ, independent of sufficiently large R, fixes S near the boundary and satisfies

$$\Phi^* g \leq \widehat{g}_R, \qquad \mathsf{A}_{\widehat{g}_R}(S) \geq \mathsf{A}_g(S). \tag{2.4}$$

This is a metric theorem in every $n \geq 3$; it needs no DEC, K, ADM limit, minimizing hypersurface, or positive area assumption.

Proof. Choose $T > 0$ with $e^{-2T} < c$, with strict slack. By the last assertion of Lemma 2.1, there is a radius R_* such that $\Phi_T^* g \leq c\delta$ for $r \geq R_*$. For $R \geq R_*$, on $r \leq R$ use $\Phi_T^* g \leq g \leq \widehat{g}_R$; on $r \geq R$ use the tail inequality. This proves the tensor comparison.

If D is an admissible outer domain, $\Phi_T(D)$ remains connected, smooth and closed, with interior in the open exterior. It contains the sufficiently distant end: the omitted compact part stays compact under this proper homeomorphism. The full intrinsic boundary is $\Phi_T(\partial D)$, retaining every component and all contact with S. For a k-plane, a positive-semidefinite metric comparison implies the corresponding determinant comparison. Hence

$$\mathrm{Area}_{\widehat{g}_R}(\partial D) \geq \mathrm{Area}_g(\Phi_T(\partial D)) \geq \mathsf{A}_g(S).$$

Take the infimum over D. No limiting minimizing sequence was needed. □

Corollary 2.3 (The source's fixed-boost replacement)**.** *Assume the source's four-dimensional rest-end construction has produced the metrics and bounds in weak4.tex, lines 637–651 and 723–751. For fixed original data and fixed future-timelike boost, the output satisfies*

$$\mathsf{A}_{g_R}(S) \geq \mathsf{A}_g(S)$$

for all sufficiently large R, replacing its $(1+\epsilon_R)^{-3/2}$ comparison. The same conclusion applies to the strong-decay replacement whenever its stated core domination and uniform lower end bound hold. Subsequent nonnegative conformal strictification preserves this inequality.

Proof. The repair factor is at least one and the intermediate metric is the original one inside R. The source gives $g_R \geq c\delta$ outside, where c depends on the fixed boost but not on large R. Its O_2 original decay implies the O_1 hypothesis. Apply Theorem 2.2. Multiplication by $e^{2s\phi} \geq 1$ can only increase every cut area. □

What this does not say. This does not establish the replacement metrics, their DEC, or their rest energy $m_R \to m$. It does not assert convergence of their area infima, equality of the two metrics, or any charge invariance under Φ. The asymptotic dilation is deliberately not an ADM coordinate identification. The constants need not be uniform as $E - |P| \downarrow 0$. Nor does this remove the factor $e^{-k\varepsilon}$ in the later filled-graph deformation, whose metric need not dominate g on the whole compact core.

2.1 A warped example and a necessary hypothesis

For $g = a(r)^2 dr^2 + b(r)^2 \sigma_k$, let a radial map be $\Phi(r,\theta) = (s(r),\theta)$. Its two tensor comparison conditions are

$$a(s)^2 s'(r)^2 \le a(r)^2, \qquad b(s)^2 \le b(r)^2.$$

For the flow above these follow from $\mathcal{L}_X g \ge 0$, not from coordinate radius alone. In the Euclidean case $a = 1, b = r$, $X = \chi(r^2/2) r\partial_r$. For fixed time,

$$\dot{s} = -\chi(s^2/2)s, \qquad \frac{d}{dt}\partial_r s = -[\chi(s^2/2) + s^2\chi'(s^2/2)]\partial_r s.$$

Thus $0 < \partial_r s \le 1$, $s \le r$, and on a far tail $s = e^{-T} r$. A metric that is Euclidean on $r \le R$, interpolates smoothly, and is $c\delta$ past $2R$, with fixed $0 < c < 1$, can have smaller coordinate-sphere areas in that far region, yet Theorem 2.2 still controls every full enclosing cut.

The fixed tail lower bound is substantive. In the Euclidean exterior $r \ge 1$, choose a smooth conformal metric equal to δ on $r \le R$, with conformal length factor R^{-2} on a neighborhood of $r = 2R$, and returning to δ farther out. The sphere there has area $\omega_k(2/R)^k$, while the original enclosing area is ω_k. A lower bound $c_R\delta$ with $c_R \to 0$ therefore does not give the theorem's conclusion. These metrics are only a metric counterexample; no DEC is asserted.

3 The actual filled-graph source and its measures

We now fix one prepared datum in the source's sense. Thus K is compactly supported, $g - \delta = O_d(r^{2-n})$ for every fixed d, the boundary is strictly future trapped, and $\mu - |J| \ge c\varrho^{-n-\beta}$ for $0 < \beta < 1$. The source fills the entire boundary by a smooth compact cap and product necks; the resulting $\mathcal{M}_N$ has no boundary and one end. No DEC is assumed in the filling.

Fix $\varepsilon > 0$ and $N > \max(k, 2/\varepsilon)$. With the source's smooth nonincreasing cutoff ϑ, put

$$p(t) = N\vartheta(Nt), \quad l(t) = \exp\left(\int_0^t p(a)\,\mathrm{d}a\right), \quad \ell = e^{-N\varepsilon}, \quad \eta = \ell^{3/2}, \tag{3.1}$$

$$m_0(t) = \vartheta(N(t+\varepsilon)), \quad D = 1 + l(t)^2|df|_g^2, \quad Z = t + \frac{\log D}{2k}. \tag{3.2}$$

The function ϑ equals one on $(-\infty, 0]$, zero on $[1,\infty)$, and takes values in $[0,1]$. The last relation determines t smoothly from Z, df: its derivative with respect to t is $1 + pv/k \ge 1$, where $v = 1 - D^{-1}$.

Write

$$w = lD^{-1/2}\nabla f, \quad d = D^{-1}, \quad \chi = \frac{kd}{k+pv}, \quad A_\chi = I - \frac{k+p}{k+pv} w \otimes w, \tag{3.3}$$

$$H^f = lD^{-1/2}\operatorname{Hess}_g f, \quad F = \operatorname{tr}_{A_\chi}(K + H^f), \quad h = \eta f, \quad \tau = \operatorname{tr}_g K, \tag{3.4}$$

$$u = e^{(k-1)t} l\sqrt{D}, \quad V = uA_\chi(k\nabla Z + K(w,\cdot)), \quad \widehat{g} = e^{2t}(g + l^2 df \otimes df). \tag{3.5}$$

Every contraction and the divergence in this section use g. In particular $u\Xi\, dV_g$, not $\Xi\, dV_{\widehat{g}}$, is the physical source measure in the energy-flux identity.

The source's locally derived scalar identity is

$$\frac{1}{2} e^{2t} R_{\widehat{g}} = \mu + J(w) + \mathcal{T} + w(F) - F\tau - u^{-1}\operatorname{div}_g V. \tag{3.6}$$

We retain this identity as [S], including its sign convention. The relevant coercivity $\mathcal{T} \ge 0$ is not assumed for an arbitrary symmetric matrix: it is the particular completed form below. At a nonzero

gradient split $K + H^f$ into transverse block T_0, mixed vector M, and axial entry j, and write $dt = (y_0, x_0)$. With $a_w = |w|$, $s_p = p + (k-1)/2$,

$$\mathcal{T} = \tfrac{1}{2}|T_0^\circ|^2 + |M + s_p a_w y_0|^2 + \left(k s_p - \frac{(k-1)^2 v}{4}\right)|y_0|^2 \tag{3.7}$$
$$+ k s_p d\left(x_0 + \frac{a_w j}{k+pv}\right)^2 + \frac{k+1}{2k}\left(F - \frac{\chi j}{k+1}\right)^2 + \frac{kd[k(k+2)+v]}{2(k+1)(k+pv)^2} j^2.$$

All coefficients are positive for $k \geq 2,\ p \geq 0,\ 0 \leq v < 1$. At $df = 0$ the invariant identity is smooth and any axis may be chosen. This completion, and the $C_k(1+N)^2$ weighted estimate derived from it, are already in the source, not new results here. The weights d, χ cannot be deleted near a vertical graph.

Choose $n/4 < b_1 < \min(n-2, n/2)$ and $\rho_0 = \varrho^{-n-\beta}$, $\rho_1 = \varrho^{-2b_1}$, $\rho = c_* \rho_0$. The specified system is

$$F = h, \qquad \operatorname{div}_g V = u\Xi, \qquad \Xi = \delta_0(\mathcal{T} + \eta a_w |df|) + \rho - m_0 C_N(\rho_0 + v\rho_1). \tag{3.8}$$

Here C_N is the chosen penalty polynomial, and $\delta_0 > 0$ is fixed and small. This is not a zero-divergence Jang coupling. For an above-floor solution $t > -\varepsilon$, the penalty is supported in $-\varepsilon < t < -\varepsilon + 1/N < 0$, where $\ell \leq l \leq e\ell$. The source proves global existence by separate scalar estimates, two homotopies and exhaustion. Replacing (3.8) by a finite inverse or a frozen Green kernel would not reproduce that proof.

4 A sharp pointwise penalty budget

The source estimates the negative part of $u\Xi$ by Young's inequality, retaining the favorable quadratic height term. The next proposition optimizes that same competition without estimating the square root by a linear function.

Proposition 4.1 (Convex cubic certificate)**.** *At a point in the penalty band put $s = l|df|$, $y = \sqrt{1+s^2}$ and $\alpha = e^{(k-1)t}$. Define the real coefficients*

$$A = \delta_0 \eta > 0, \quad B = l\{m_0 C_N(\rho_0 + \rho_1) - \rho\}, \quad C = l m_0 C_N \rho_1 \geq 0 \tag{4.1}$$

and $H(y) = A(y^2 - 1) - By + C/y$, $y \geq 1$. Then

$$u\Xi \geq \alpha H(y) \geq -\alpha \mathcal{B}(A, B, C), \qquad \mathcal{B} = [-\min_{z \geq 1} H(z)]_+. \tag{4.2}$$

The minimum is uniquely attained at $y_* = 1$ if $B + C \leq 2A$; otherwise $y_* > 1$ is the unique solution on $(1, \infty)$ of

$$2A y_*^3 - B y_*^2 - C = 0. \tag{4.3}$$

At an interior minimizer,

$$H(y_*) = -A(y_*^2 + 1) + 2C/y_*. \tag{4.4}$$

This is the exact minimum of the retained scalar expression, not a claim that a global PDE solution realizes the minimizing slope.

Proof. Use $u = \alpha l y$, $v = (y^2 - 1)/y^2$, and

$$u\eta a_w |df| = \alpha \eta l^2 |df|^2 = \alpha \eta (y^2 - 1).$$

Dropping only $u\delta_0\mathcal{T} \geq 0$ gives precisely $\alpha H(y)$. Also

$$H'(y) = 2Ay - B - C/y^2, \qquad H''(y) = 2A + 2C/y^3 > 0.$$

Thus H is strictly convex and coercive. Its derivative at one determines whether the minimum is at the endpoint; otherwise the derivative has exactly one zero, yielding (4.3). Substitution of $B = 2Ay_* - C/y_*^2$ proves (4.4). This also handles $C = 0$ without division by C. Outside the penalty band $m_0 = 0$, all terms of Ξ are nonnegative. □

Proposition 4.2 (Rational bracketing and a simple upper certificate)**.** *If $B + C > 2A$, a valid upper bracket is*

$$U = 1 + \frac{B_+ + C}{2A}, \qquad 1 < y_* \leq U.$$

Given $1 \leq a \leq y_* \leq b$, convexity gives, for every $z \in [a, b]$,

$$H(z) - |H'(z)|(b - a) \leq H(y_*) \leq H(z). \tag{4.5}$$

Consequently rational sign bisection of H' supplies two-sided certificates for $\mathcal{B}$ whenever A, B, C are rational. Independently of root computation,

$$\mathcal{B} \leq lm_0C_N\rho_0 + \frac{l^2m_0^2C_N^2(\rho_0 + \rho_1)^2}{4\delta_0\eta}. \tag{4.6}$$

Proof. At U, $2AU \geq B + C + 2A$, while $C/U^2 \leq C$, so $H'(U) > 0$. The tangent inequality $H(y_*) \geq H(z) + H'(z)(y_* - z)$ proves (4.5). All denominators in the derivative are positive, so its rational sign is determined exactly by the cubic numerator. For (4.6) discard the positive ρ term, use $\sqrt{1 + s^2} \leq 1 + s$ and $s^2/\sqrt{1 + s^2} \leq s$. The remaining expression is at least

$$\delta_0\eta s^2 - lm_0C_N(\rho_0 + \rho_1)s - lm_0C_N\rho_0.$$

Complete this quadratic square and take its infimum. □

The exact pointwise minimization never gives a worse bound than (4.6). It retains the positive source ρ and the geometric v-weight; those are both lost in the simple majorant. For a manufactured local fixture take

$$l = m_0 = C_N = \rho_1 = 1, \qquad \rho_0 = 3, \quad \rho = \tfrac{1}{4}, \quad \delta_0\eta = 1.$$

Then $A = C = 1$, $B = 15/4$, $y_* = 2$, and $\mathcal{B} = 4$. The simple majorant is $3 + 16/4 = 7$. The fixture tests the local algebra only; it is not a solution of the source equations or a value achieved on its penalty band.

Corollary 4.3 (A refined ADM upper budget)**.** *Assume a smooth above-floor solution has the source's fixed-N end asymptotics $f = O_j(e^{-c_N r})$, $t = O_j(r^{2-n})$ and $\Delta_g t = O(r^{-n-\beta})$, and all terms are integrable as stated there. Let $\mathcal{P}$ denote its penalty band. Then*

$$E(\widehat{g}) - E(g) \leq \frac{1}{k\omega_k} \int_{\mathcal{P}} e^{(k-1)t}\mathcal{B}(A, B, C)\, \mathrm{d}V_g. \tag{4.7}$$

The right side is bounded by a fixed polynomial times $\ell + \ell^2/\eta = \ell + \ell^{1/2}$, but its directly evaluated value can be smaller.

Proof. On the complete filling the divergence theorem has no inner flux. With the stated asymptotics, $V = k\nabla t + o(r^{1-n})$ in integrated flux, and the graph term contributes zero ADM flux. The conformal linear term $2t\delta$ contributes $-2k\,\partial_i t$, so

$$E(\widehat{g}) - E(g) = -\frac{1}{k\omega_k}\int_{\mathcal{M}_N} u\Xi \,\mathrm{d}V_g.$$

Apply Proposition 4.1. In the band, $e^{(k-1)t} \le 1$, $m_0 \le 1$, and $l \le e\ell$. Equation (4.6) is therefore dominated by

$$e\ell C_N \rho_0 + \frac{e^2\ell^2 C_N^2}{2\delta_0\eta}(\rho_0^2 + \rho_1^2).$$

The source's filling has core volume $O(1+N)$; the tail integrals converge because $\rho_0 \in L^1$ and $4b_1 > n$. This reproduces the source's asymptotic order with explicit scalar majorants. It is not an improvement of the exponent $1/2$. □

5 Trace, divergence and boundary residuals

The private original PDE-first proofs [2, 3] distinguish weak source dual norms, actual conormal data, and nonlinear coefficient variation. Here that distinction has an especially concrete form: the trace error enters scalar curvature after a directional derivative, whereas the divergence error enters the ADM budget as a signed physical integral.

Proposition 5.1 (Exact residual transport)**.** *Let f, Z be smooth trials with the definitions (3.1)–(3.8), but not necessarily solving the equations. Put*

$$e = F - h, \qquad r = \operatorname{div}_g V - u\Xi.$$

Then the exact identity is

$$\begin{aligned}\tfrac{1}{2}e^{2t}R_{\widehat{g}} = \mu + J(w) - \rho - h\tau + (1-\delta_0)(\mathcal{T} + \eta a_w|df|) \\ + m_0 C_N(\rho_0 + v\rho_1) + w(e) - \tau e - r/u.\end{aligned} \tag{5.1}$$

On a region where the first line and penalty term have lower bound $b(x) \ge 0$, nonnegative scalar curvature follows from the sufficient pointwise condition

$$|w(e)| + |\tau e| + (r/u)_+ \le b. \tag{5.2}$$

If the trial also has the end asymptotics of Corollary 4.3, $r \in L^1(dV_g)$, and the filled divergence theorem applies, then

$$E(\widehat{g}) - E(g) \le \frac{\displaystyle\int_{\mathcal{P}} e^{(k-1)t}\mathcal{B}\,dV_g + \int_{\mathcal{M}_N} r_-\,dV_g}{k\omega_k}. \tag{5.3}$$

Proof. In (3.6) substitute $F = h + e$ and $u^{-1}\operatorname{div} V = \Xi + r/u$. Since $w(h) = \eta a_w|df|$, the terms collect exactly as (5.1). Bounding only the potentially negative residual terms proves (5.2). Integrating $\operatorname{div} V = u\Xi + r$, with the same ADM calculation as before, gives the energy change $-(k\omega_k)^{-1}\int(u\Xi + r)dV_g$. Apply the lower bound for $u\Xi$ and retain the negative part of r. □

The signs in these two budgets are different. A positive divergence residual lowers ADM energy but can harm scalar curvature; a negative one costs ADM energy but improves the scalar identity. Thus a single unsigned algebraic residual does not express both mechanisms. One may use $|w| \le 1$ to replace $|w(e)|$ by $|de|_g$, but a small L^2 norm of e alone supplies neither bound.

On a fixed smooth bounded truncation $U \subset \mathcal{M}_N$, a divergence equation with $V \in L^2$, $u\Xi \in L^2$ has weak residual

$$\mathcal{R}(\varphi) = -\int_U \langle V, d\varphi\rangle_g \, dV_g - \int_U u\Xi\varphi \, dV_g, \quad \varphi \in H_0^1(U).$$

This is an H^{-1} functional. It does not determine the outer flux without additional trace information. For example on a Euclidean ball, $V = x/n$ has $\operatorname{div} V = 1$, and the correct weak equation with source one has zero interior residual while its outer flux is $\mathrm{Vol}(U)$. An interior residual does not encode a zero-flux boundary condition. For the source construction the outer data are $f = Z = 0$ on exhaustion spheres; the filling has no inner boundary. Restricting instead to the original exterior creates an actual inner boundary term, with normal opposite to the convention for θ_+. It cannot be dropped.

The private Green equivalence theorem requires a specified invertible weak operator with the correct trace space. Here the coefficients A_χ, u depend on f, Z, the trace equation is nonlinear, and its axial eigenvalue can be small. A frozen finite matrix can evaluate a trial but cannot supply a global inverse or continuation theorem. Our residual proposition assumes the trial geometry and its asymptotics; it does not manufacture those hypotheses.

6 Finite-stage interpretation and usable area lower bounds

An exactly transported area is still an infimum over infinitely many cuts. A useful classical lower certificate, consistent with the source's tube proof of positivity, is as follows.

Proposition 6.1 (A flux certificate with a defect)**.** *Let ω be a smooth k-form on the one-ended exterior, smooth up to S, with g-comass at most $M > 0$. Suppose $d\omega = f\, dV_g$, $f \in L^1(dV_g)$, and the oriented outer flux has limit $\lim_{R\to\infty} \int_{S_R} \omega = Q$. Then every full cut obeys*

$$\mathrm{Area}_g(\Gamma) \ge \frac{(|Q| - \|f\|_{L^1})_+}{M}. \tag{6.1}$$

In particular this lower bound applies to $\mathsf{A}_g(S)$, and is inherited by the metrics of Theorem 2.2.

Proof. Orient Γ toward the retained end. For sufficiently large R, Stokes on the compact truncated outer domain gives

$$\int_{S_R} \omega - \int_\Gamma \omega = \int_{D\cap\{r\le R\}} f \, dV_g.$$

The outward boundary orientation on the inner cut is the negative of the chosen one; it includes all components, even those on S. Let $R \to \infty$, using absolute integrability, and bound $|\int_\Gamma \omega| \le M \,\mathrm{Area}_g(\Gamma)$. Taking infima proves the result. No sampling of cuts occurs. □

This is Stokes' theorem with comass control, not a new calibration principle. Its role is to name the global data that a lower-area certificate must possess. A small closedness error only in a compact region, with an uncontrolled end tail, does not give (6.1). For Euclidean $r \ge 1$, the pulled-back unit sphere area form has $Q = \omega_k$, $d\omega = 0$, and comass $r^{-k} \le 1$. The boundary sphere attains the resulting lower bound, so $\mathsf{A}_\delta = \omega_k$. This justifies the counterexample in Section 2.1.

A transparent flow threshold. On a Euclidean radial region $r \geq R_0 > 0$, the increasing profile $\chi(F) = 1 - R_0^2/(2F)$ has the explicit negative-time flow

$$s_T(r)^2 = R_0^2 + z(r^2 - R_0^2), \qquad z = e^{-2T} \in (0,1).$$

Its squared tangential and radial dilations are respectively

$$z + (1-z)R_0^2/r^2, \qquad \frac{z^2}{z + (1-z)R_0^2/r^2}.$$

Both are at most one. If $z < c < 1$, both are at most c on

$$r^2 \geq \frac{(1-z)R_0^2}{c-z}.$$

For instance $z = 1/4,\ c = 1/2$ gives $r \geq \sqrt{3}R_0$. This explicit profile is a smooth exterior example fixing its boundary, not the flat-on-a-collar cutoff of Lemma 2.1; the latter is used in the source application.

The finite prepared comparison. Assume, separately, that the source's detached Riemannian exterior has actually been constructed and that its receiving Riemannian Penrose theorem applies. If its full frontier area is at least $e^{-k\varepsilon}A_{\mathrm{low}}$ and its energy is at most $E_0 + B_N$, then

$$E_0 \geq \frac{1}{2}e^{-(n-2)\varepsilon}\left(\frac{A_{\mathrm{low}}}{\omega_k}\right)^{(n-2)/k} - B_N. \tag{6.2}$$

Here B_N can be the integral in (4.7), since the source's further plateau factor is identically zero near infinity and does not change energy. For approximate trials it can instead be (5.3) only if all the other geometric hypotheses, including curvature and detachment, have been verified independently. Equation (5.2) by itself is not a detachment theorem.

This finite inequality is an exact rearrangement of conditional geometric bounds. For a desired relative area factor $1 - \delta$, first choose a fixed $\varepsilon > 0$ with $e^{-(n-2)\varepsilon} \geq 1 - \delta$, then bound B_N for that fixed prepared datum and fixed ε. The source's polynomial constants may depend on ε; choosing $\varepsilon = \varepsilon(N)$ without new uniform estimates is not licensed. Likewise exhaustion precedes $N \to \infty$; strictification is removed at a fixed rest-replacement radius, which is sent to infinity last. The exact cut comparison removes one area loss but none of these analytic orders of limits.

7 Dependency map, comparison, and limitations

7.1 Source architecture

The assigned source, rather than any neighboring family report, was used. Its scientific modules have the following roles.

Module	Mathematical role and status in this report
introduction	Data, full-cut definition, strong and weak decay claims; read in full.
prepared-comparison	Exact receiving frontier hypotheses and numerical limit order; read in full. The companion endpoint remains imported.
end-preparation	Matching charges, compact constraint corrections, vacuum bending, DEC repair and strictification. Opening conformal laws and full-cut comparison read; other proofs imported.
scalar-identity	Filled graph, exact scalar identity, weighted coercivity, floor and penalty flux. Read in full; selected identities rederived locally, global use conditional.
elliptic	Height and upper barriers, scalar regularity, compact map, homotopies and exhaustion. Opening theorem and final exhaustion/end proof read; intervening global analytic proof not audited.
height-enclosure	Height separation, conformal plateau, two densities, BV detachment, complete full frontier and original area. Read in full; GMT and global analytic inputs retained.
weak3	Different physical density convention and enclosing class; first 201 lines read. Remaining weak preparation imported.
weak4	Conformal laws, rest replacement, all-cut comparison, strictification, numerical transfer and momentum flux. Read in full.

Reading an argument is not an independent proof certification of all its external hypotheses. In particular we have not supplied the compact constraint right inverses, the nonlinear filled-system continuation, singular-frontier conformal flow, or rigidity at equality. The source's designated-end four-dimensional class excludes every other distant end; the three-dimensional complete-data class is different. Our one-ended theorem does not identify those two infima.

7.2 What the Ram comparison contributes

The fresh private readings were the actual local originals [2, 3], not earlier M-series comparison reports. Their useful principles are exact weak source/boundary specification, retaining coefficient derivatives, and distinguishing true dual residual bounds from finitely tested residuals. Those principles are implemented in Proposition 5.1; the constants, gradient trace error and $u\,dV_g$ weight are determined by this source, not copied from a generic Green problem.

The cutoff convex-gradient flow and the optimized penalty expression are the report's local deductions. Lie differentiation, ODE continuation, determinant monotonicity, Stokes and convex minimization are classical tools. Neither these tools nor the source's scalar identity and coercivity are credited as newly invented Ram results. No claim is made that all prior literature lacks either local lemma.

7.3 Advantages, costs, and failure modes

Exact area transport avoids a minimizing-cut compactness theorem and one asymptotic area factor; it applies even to disconnected cuts and boundary contact. The cost is actual C^1 asymptotic control and a uniform tail lower bound in the original chart. It cannot be inferred from a finite sample of metric matrices. The gradient flow depends on the original geometry, and does not directly compute the area infimum.

The cubic certificate retains positive forcing and the precise graph weights. It admits monotone scalar bracketing and can outperform a coarse Young bound, but evaluating its integral on an actual solution requires controlled fields and quadrature/tail errors. Small values on a finite grid alone are not an integral upper certificate. The polynomial exponential rate is unchanged.

Residual transport explains separately which sign costs curvature and which costs mass. Its pointwise curvature test is more demanding than an H^{-1} test, and the end flux needs genuine integral and trace control. A formal Green inverse or invertible finite Jacobian is not an existence proof for the source's nonlinear exterior system.

7.4 Feasible extensions

The flow proof works for any finite collection of ends on which the same hypotheses and end-preserving cut class are explicitly imposed: choose disjoint end cutoffs and sum their vector fields. This geometric extension does not give a multi-end Penrose theorem. A further perturbative problem is to replace exact core domination by a certified tensor defect and quantify its effect on every tangent k-plane. Without localization of all competing cuts, a small integrated metric defect is insufficient.

For the analytic construction, one can seek interval enclosures for $t, df, \rho, \mathcal{T}$, validate the cubic budget with physical volume weights, and retain the signed divergence source. To promote such computations to PDE existence would additionally require infinite-dimensional inverse or continuation bounds, exact boundary conditions and admissible end decay. No such validation is reported here. Finally, uniform control in ε, the boost, or the preparation radius would be a separate research task, not a consequence of exponential decay at fixed parameters.

8 Current literature and verification scope

The public refresh was performed on October 9, 2026 using public titles and identifiers only. Allen, Bryden, Kazaras and Khuri [4] give a suboptimal-constant spacetime bound with their three-dimensional enclosing-horizon setting. Ellithy [5] uses a quasi-final-state hypothesis on a spacetime horizon tube. These are different statements, not public confirmation of all the assigned source's arbitrary-dimensional assertions. The abstract and version records inspected show v1 dated April 14, 2025 and May 18, 2026 respectively.

Bray–Khuri [6] explicitly distinguish their spherical theorem from the desired nonsymmetric coupled-system existence theory. Jaracz [7] gives a radial nonexistence obstruction for a specified generalized-Jang/zero-divergence coupling. That obstruction does not by itself refute the different sourced system (3.8); conversely changing the source is not an existence proof. For these external papers the abstract/publisher metadata were inspected, not their complete proofs.

Literal-read scope. The owned ledger records 3,812 freshly displayed public-source lines and 459 private-proof lines. The complete introduction, comparison, scalar-identity, height-enclosure and weak4 modules were displayed in bounded untruncated chunks. Additional public reads were end-preparation 1–98 and 573–627, elliptic 1–89 and 1567–1697, weak3 1–201, and the root, preamble, README and figure files. The bibliography file was inventoried and hashed, not literally read. The private passages were TR8 lines 81–195, 326–394, 534–578 and TR1 stability lines 1–230. Exact unread complements, pre-read capture time, post-read hashes and unchanged status are in the owned ledger. Headings-only searches are discovery, not proof reading. No whole-corpus or whole-public-manuscript proof audit is claimed.

Owned checks. The companion executable uses exact rational arithmetic for manufactured radial-flow dilations, Schwarzschild-type Hessian eigenvalues, cubic minima and brackets, penalty source identities, signed residual recombination and dimension-dependent normalizations. It records

parameter cases separately from individual assertions. The executed suite passed 2,057 exact parameter cases and 9,291 assertions; there were no floating-point test cases. No foreign checker or parent test total is counted. These fixtures test local formulas and degeneracies, not actual initial data satisfying all source premises, an infinite set of cuts, or global elliptic solvability. Final executed counts are recorded in the check receipt accompanying this report.

9 Conclusion

The strongest increment is an exact full-cut inequality under the source's already stated end-replacement metric bounds. It follows from a global proper contracting map, not from replacing physical area by a coordinate radius. The second increment is a sharp one-dimensional adverse-source budget with the correct graph volume weight and ADM sign. Both have complete local proofs and explicit limitations. The global spacetime Penrose construction remains upstream, and applicability L3 does not itself justify contribution C3; the latter is assigned only to these demonstrated refinements.

References

[1] OpenAI, *The spacetime Penrose inequality and enclosing area*, September 27, 2026. Cached source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Pinned public manuscript. The captured October 8 tree has identical source bytes according to the parent identity record; this is provenance, not proof certification.

[2] M. Subbarao, private PDE-first Ram corpus, original local `RamNonlin-IE-PDE-astra`, Report 8: `TR8-lin-non-lin.tex`. Fresh passages specified in Section 8. Confidential source retained locally.

[3] M. Subbarao, private Ram integral-equation corpus, Report 1, *Perturbation, regularization, and a posteriori control*, local `TR1-lin-nonlin/stability.tex`, lines 1–230. Confidential source retained locally.

[4] B. Allen, E. Bryden, D. Kazaras and M. Khuri, *Proof of the Spacetime Penrose Inequality With Suboptimal Constant in the Asymptotically Flat and Asymptotically Hyperboloidal Regimes*, arXiv:2504.10641v1 (2025). Abstract and version metadata accessed October 9, 2026; full proof not independently audited here.

[5] A. Ellithy, *The spacetime Penrose inequality under a quasi final state hypothesis*, arXiv:2605.18730v1 (2026). Abstract and version metadata accessed October 9, 2026; full proof not independently audited here.

[6] H. L. Bray and M. A. Khuri, *A Jang equation approach to the Penrose inequality*, Discrete Contin. Dyn. Syst. 27 (2010), 741–766. Publisher record and abstract, accessed October 9, 2026.

[7] J. S. Jaracz, *Nonexistence of Solutions to the Coupled Generalized Jang Equation/Zero Divergence System*, arXiv:2304.09332v1 (2023); Class. Quantum Grav. 40, 195013. Abstract and metadata accessed October 9, 2026.